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A simple lattice Boltzmann model for turbulence Rayleigh–Bénard thermal convection

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ABSTRACT

A simple lattice Boltzmann thermal model is proposed for simulating the turbulence Rayleigh–Bénard convection. This new model introduces a mesoscopic external force term in the lattice Boltzmann equation to consider the effect of the buoyant body force and introduces a temperature density distribution function to simulate the temperature field. The macroscopic density and velocity fields are simulated using the density distribution function. The results show that this simple model is able to simulate turbulence Rayleigh–Bénard convection in the current simulation range of $10^6 \leqslant Ra \leqslant 10^{12}$. It is shown that the energy spectra of the velocity and temperature are in good agreement with the Bolgiano–Obukhov-like (*BO59*) scaling.

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1. Introduction

In the past few years, turbulent convection has been the subject of extensive studies [1,2]. Several fundamental studies deal with the Rayleigh–Bénard problem in order to gain deeper knowledge of underling the mechanism driving the convective flow and the heat transport [2]. Zhou and Xia's [3] experiments show that the skewness of the temperature signal can be used to separate the flow field into three regions: bulk, boundary and mixing layers. The influence of an isothermal and a constant heat flux bottom plate are investigated on the heat transport in a cylindrical geometry. It is indicated that the unsteady ejection of thermal plumes with a constant heat flux boundary is closer to experimental conditions [2]. The thermal plumes are, however, a subject area that attracted a lot of attention in the field of thermal convection. Shishkina and Wagner [4–6] also performed direct numerical simulations (DNS) in a cylindrical geometry and analyzed the geometrical properties of sheet-like thermal plumes in horizontal slices through the bulk flow.

The lattice Boltzmann method (LBM) as a numerical scheme has achieved considerable success in simulating fluid flows and associated transport phenomena [7,8], ranging from low-Reynolds-number flows to highly turbulent flows. Calculations with LBM based on Direct Numerical Simulations (DNS), Large Eddy Simulations (LES), and Reynolds-Averaged Navier–Stokes (RANS) were implemented, which showed great success in turbulence predictions [9]. LBM is supposed to be superior to conventional Navier–Stokes (N–S) equation based computational fluid dynamics (CFD) method on turbulent flow [9]. First of all, LBM is able to present rich and complex physics when projected onto hydrodynamic variables in x space. Second, both turbulent and thermal fluctuations are coupled at the same level. Third, the parameter-relaxation time, τ, naturally includes both the regular viscous effects and its higher order modifications.

In general, LBM has not yet extensively used in simulations of thermal systems, compared to that of isothermal flows. Theoretically, a LBM model with energy conservation can be constructed to yield a temperature evolution equation at the macroscopic level. However, the application of LBM to heat transfer problems has not achieved great success for the thermal models due to the severe numerical instability caused by breaking the isothermal [10]. Various numerical simulations have been performed using different thermal lattice Boltzmann (LB) models or Boltzmann-based schemes to investigate Rayleigh–Bénard convection [11,12]. Although the results provided by these studies for stationary convection are in good agreement with the data [12]. In the literature, the thermal LBM models were used to simulate Rayleigh–Bénard thermal convection in the range of $Ra \leqslant 10^6$ [12]. However, when Ra is higher than 10^6, the scheme of these thermal LBM models is numerically unstable. The previous thermal LBM models are not able to simulate turbulence Rayleigh–Bénard convection [12].

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In this paper, an external force is introduced to the lattice Boltzmann equation. This new model introduces a temperature density distribution function to simulate the temperature field. The macroscopic density and velocity fields are simulated using the traditional density distribution function. This simple model based on DNS is able to simulate turbulence Rayleigh–Bénard convection in the range of $10^6 \leqslant Ra \leqslant 10^{12}$. A good agreement between the LBM and the other methodologies has been observed, confirming that the method provides an accurate description of the fluid momentum and temperature fields. In the following, a simple lattice Boltzmann thermal model will be briefly described at first. Then the problem definition and numerical verification are followed. After that, the detailed numerical study of turbulence Rayleigh–Bénard convection is given. Finally, some concluding remarks are provided.

2. Numerical method

The numerical algorithm used is based on discrete kinetic models [11]. The starting point is a standard coupled mesoscopic dynamics described by [10,11]:

$$f_i(x + c_i\Delta t, t + \Delta t) = f_i(x,t) + (f_i^{eq}(x,t)) - (f_i(x,t))/\tau_v + F_i \tag{1}$$

$$g_i(x + c_i\Delta t, t + \Delta t) = g_i(x,t) + (g_i^{eq}(x,t)) - (g_i(x,t))/\tau_T \tag{2}$$

where $f_i(x,t), g_i(x,t)$ stand for the probability density functions to find at (x,t), c_i is a discrete particle velocity (with $i = 0, 1 \ldots 8$ in the $D2Q9$ LBM adopted here [13]), F_i is a mesoscopic force according to the buoyant body force, τ_v and τ_T are the relaxation time for flow and temperature LB equations, respectively. In this study, an expression for mesoscopic discrete force is proposed for simulating turbulence Rayleigh–Bénard convection problem. The additional buoyant body force term, F_i, can be formulated by the Boussinesq approximation, i.e.

$$F_i = 3w_i \cdot \rho \cdot \mathbf{g} \cdot \beta \cdot T \cdot c_{iy} \tag{3}$$

where $\mathbf{g}$ is the acceleration of gravity, β is the thermal expansion coefficient, and c_{iy} is the y-component of c_i.

Density, momentum, and temperature are defined as coarse-grained fields of the distribution functions

$$\rho = \sum_{i=0}^{8} f_i \quad \rho u = \sum_{i=0}^{8} c_i f_i, \quad T = \sum_{i=0}^{8} g_i \tag{4}$$

The local kinetic equilibria $f_i^{eq}(x,t)$ and $g_i^{eq}(x,t)$ are expanded in a polynomial basis [10] such that a Chapman–Enskog expansion leads to the equations for density, momentum, and temperature (Eqs. (1) and (2)): the streaming step on the left-hand side reproduces the inertial terms in the hydrodynamical equations, whereas dissipation and thermal diffusion are connected to the relaxation (toward equilibrium) properties in the right hand side, with τ_v and τ_T related to the relaxation times [8]. Thus, the following equations, can be obtained,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{5}$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot (2\rho\nu\mathbf{S}) - g\beta\Delta T \tag{6}$$

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \Gamma \nabla^2 T \tag{7}$$

The following section is the introduction of *Rayleigh* number in the Rayleigh–Bénard convection. Firstly, the critical *Rayleigh* number, Ra_c is calculated, at which the static conductive state becomes unstable, where the *Rayleigh* number is defined as

$$Ra = \frac{\beta \Delta T g H^3}{\nu \Gamma} \tag{8}$$

where g is the acceleration due to gravity, ΔT is the vertical temperature difference across the fluid layer of height H, and β, ν and Γ are, respectively, the thermal expansion coefficient, kinematic viscosity and thermal diffusivity of the fluid.

The initial condition of simulation is started from a perturbed static conductive state. The initial temperature field is given by the initial pressure field [9]. Once the Rayleigh–Bénard convection is established, the heat transfer between the top and bottom walls is greatly enhanced. The enhancement of the heat transfer can be calculated by the *Nusselt* number

$$Nu = 1 + \frac{\langle u_y T \rangle}{\Gamma \Delta T / H} \tag{9}$$

where u_y is the vertical velocity, ΔT is the temperature difference between the bottom and top walls, H is the channel height, and $\langle \cdot \rangle$ represents the average over the whole flow domain.

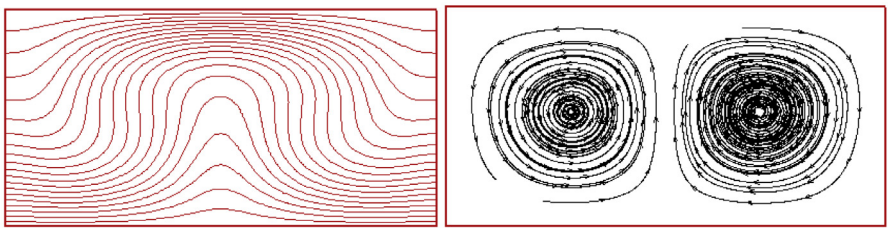
3. Numerical simulations and discussions

The first case of calculations is two-dimensional channel. Simulation at high *Rayleigh* numbers is performed on a 800×400 lattice with a *Prandtl* number of 0.71. The simulation is started from the static conductive state, beginning with $Ra = 10{,}000$. After the steady state is reached, the *Rayleigh* number is raised step by step to higher value. The side boundaries of the channel are assigned periodic boundary conditions for both the flow and the temperature fields. Meanwhile, the upper and bottom solid boundaries are assigned no-slip conditions for the flow field and are assigned constant temperatures of $T_{up} = 0$ and $T_{down} = 1$.

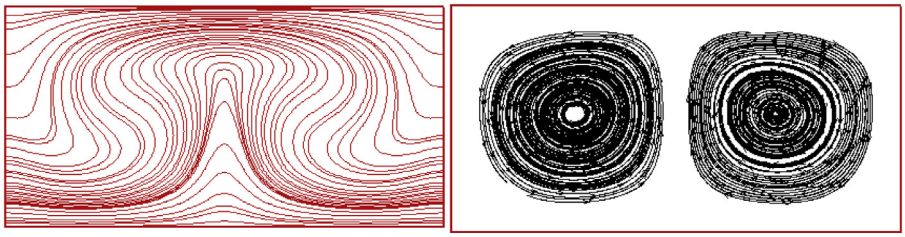
The calculated typical temperature distributions and flow patterns of Rayleigh–Bénard convection at final steady states are plotted in Fig. 1 for $Ra = 5000$, 10,000, 50,000 and 100,000. As is shown, hot fluid near the bottom wall flows upward which increases the temperature in the central portion of the channel, while cold fluid near the top wall flows downward which decreases the temperature near the side boundaries. When the *Rayleigh* number increases, two trends are observed for the temperature distribution: enhanced mixing of the hot and cold fluids and resulted in an increase in the temperature gradient near the bottom and top boundaries.

The second case of calculations is two-dimensional cubic cavity. The computational cavity is mapped by the square lattice for D2Q9 model, comprising 1000×1000 grids. This grid is also applied to simulate the Rayleigh–Bénard convection using LBM at $Pr = 0.71$. The wall boundaries of the caivity are assigned no-slip conditions for the flow field and are assigned constant temperatures of $T_{up} = 0$ and $T_{down} = 1$.

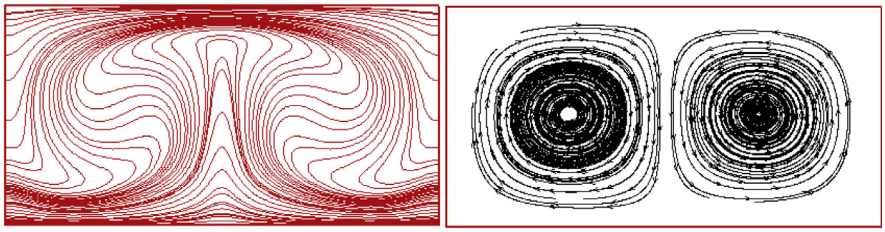
Fig. 2 presents snapshots of the dimensionless temperature fields. The red is hot plumes and the blue is cold plumes. Interactions between hot and cold plumes are presented in Fig. 2. They illustrate the formation of the thermal plumes for the three *Rayleigh* numbers spanning about four orders of magnitude from the onset of turbulence until a fully developed turbulence. At the lowest *Ra*, two pairs of plumes are rising and falling, and hence forming the typical convection rolls with their rotational axes orthogonal to the longitudinal direction of the cell. When the *Rayleigh* number is equal to 4.0×10^7, it is seen that the flow of fields gradually becomes turbulent convection and the isosurfaces of the temperature field are interrupt ted many parts. The interaction of the thermal plumes causes a large scale circulation of the flow. However, at this Rayleigh number the large-scale convection rolls are not continuously maintained, but appear to become unstable, which temporarily leads to a breakdown of the large-scale flow. However, the temperature field is entirely



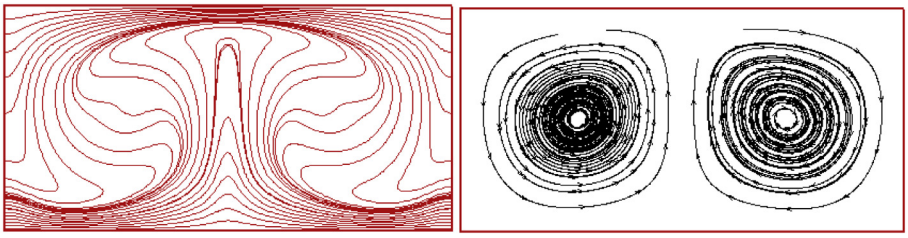
(a) Ra =5000



(b) Ra =10000



(c) Ra =50000



(d) Ra =100000

Fig. 1. Isotherms at steady states in left side and streamlines in right side as the *Rayleigh* number is raised gradually to (a) 5000, (b) 10,000, (c) 50,000 and (d) 100,000.

dominated by small-scale structures at $Ra = 1.0 \times 10^{10}$, so that a large-scale convection roll can hardly be identified in the temperature field.

The relations of the turbulent heat flux and the kinetic energy to the mean temperature gradient in their dimensionless form, i.e., the *Nusselt* number are discussed in next section. Although the existence of the ultimate state is still an open issue for the traditional Rayleigh–Bénard convection with solid boundaries, it has been shown, both numerically [6] and experimentally [2], that the ultimate regime scaling can be realized when the solid boundaries are absent. Fig. 3 shows the *Nusselt* number (Nu) as a function of *Rayleigh* number (Ra). The solid line curve is the empirical results, the blue five-pointed star represent the results of LBM and the red circles represent the simulation results of DNS by Kaczorowski [6]. As shown in Fig. 3, a clear scaling can be seen for both Nu and Ra for nearly four decades from $10^6 \leqslant Ra \leqslant 10^{10}$ and LBM simulations are in good agreement with the empirical results and DNS by Kaczorowski [6].

Fig. 4 represents kinetic energy and temperature spectra in four positions at $Ra = 3.5 \times 10^9$. Fig. 4(a) is the X direction of the kinetic energy spectrum, Fig. 4(b) is the Y direction of the kinetic energy spectrum and Fig. 4(c) is temperature spectra.

Next, the Bolgiano–Obukhov-like (BO59) scaling [14] in Fourier space will be discussed according to the simulation results. The corresponding spectra for the kinetic energy and the thermal fluctuations are [14],

$$E_u(k) \sim \varepsilon^{2/3} k^{-5/3} F(\eta k) \tag{10}$$

and

$$E_T(k) \sim (\beta g)^{-2/5} \varepsilon_T^{4/5} k^{-7/5} \tag{11}$$

where $F(\cdot)$ is a universal dimensionless function of a dimensionless argument.

The kinetic energy and temperature spectra with $Ra = 1.21 \times 10^8$ are showed at points (x/L, y/H) to (0.5, 0.5) (0.25, 0.5), (0.25, 0.98) and (0.75, 0.98). Spectra of the velocity and temperature fields extracted from the central position and averaged in time are plotted in Fig. 4. It can be seen that the velocity and temperature spectra have a roll-off rate −5/3 and −7/5, respectively, consistent with the *BO59* scaling Eqs. (10) and (11).

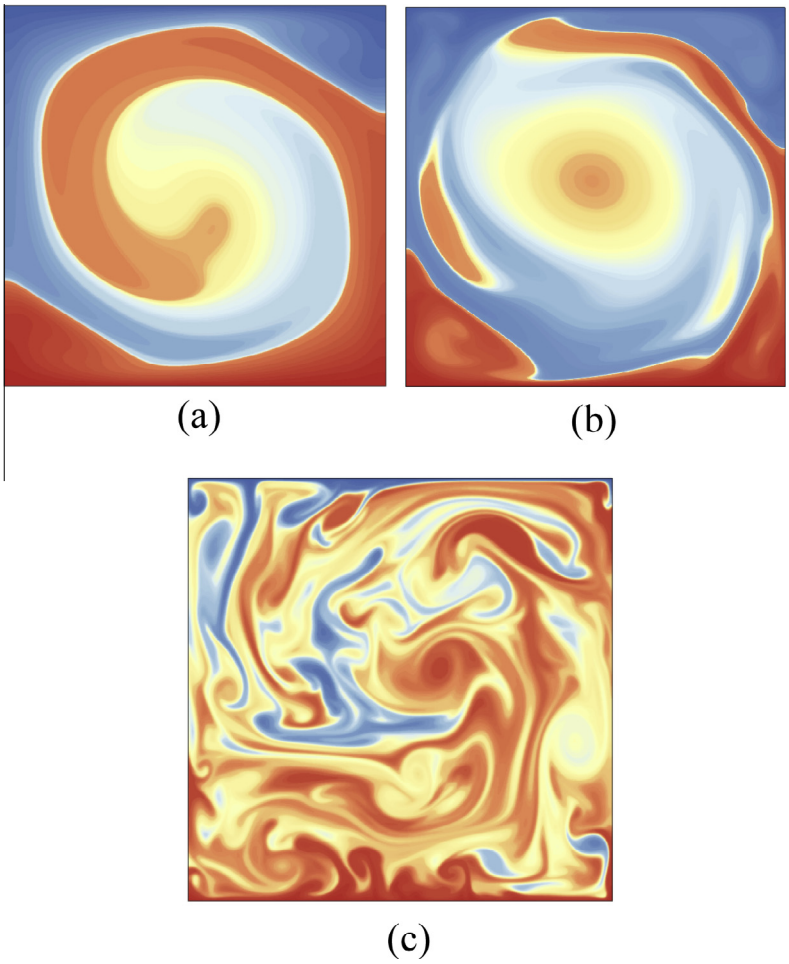


Fig. 2. Snapshots of instantaneous isotherms at $Ra = 2.0 \times 10^6$ (a), $Ra = 4.0 \times 10^7$ (b) and $Ra = 1.0 \times 10^{10}$ (c).

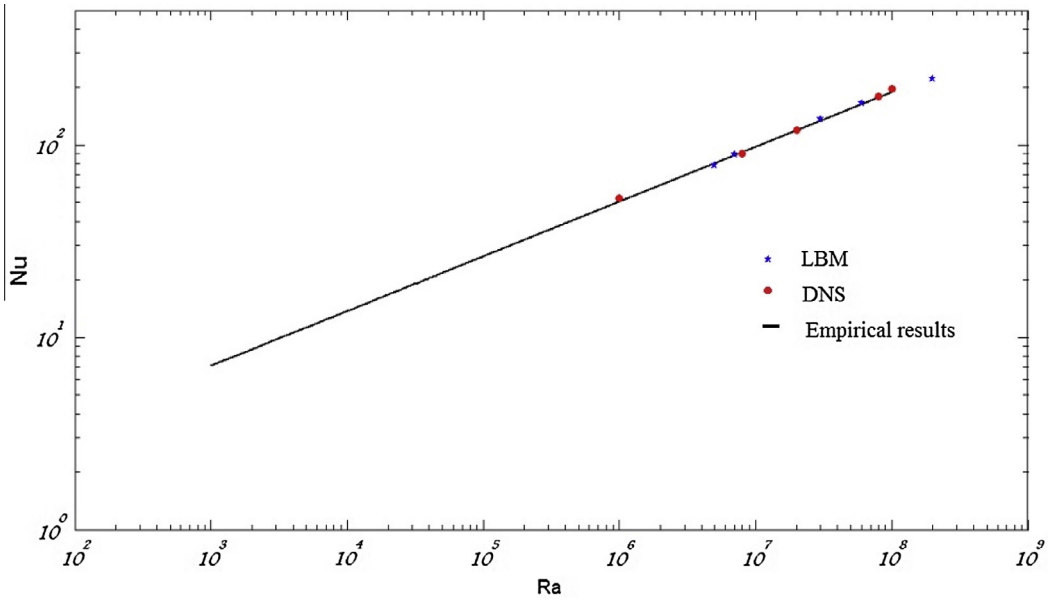


Fig. 3. *Nusselt* number as a function of *Rayleigh* number; present simulation of LBM (★) and DNS (●) adapted from by Kaczorowski [6]; $Nu \sim Ra^{0.284}$ (–).

In additions, these spectra collapse well on top of each other in the intermediate range of wavenumbers, corresponding to the scales in the inertial range, and the growth of the integral scale for both velocity and temperature at small wavenumbers is well reproduced by the temporal evolution of the compensated spectra.

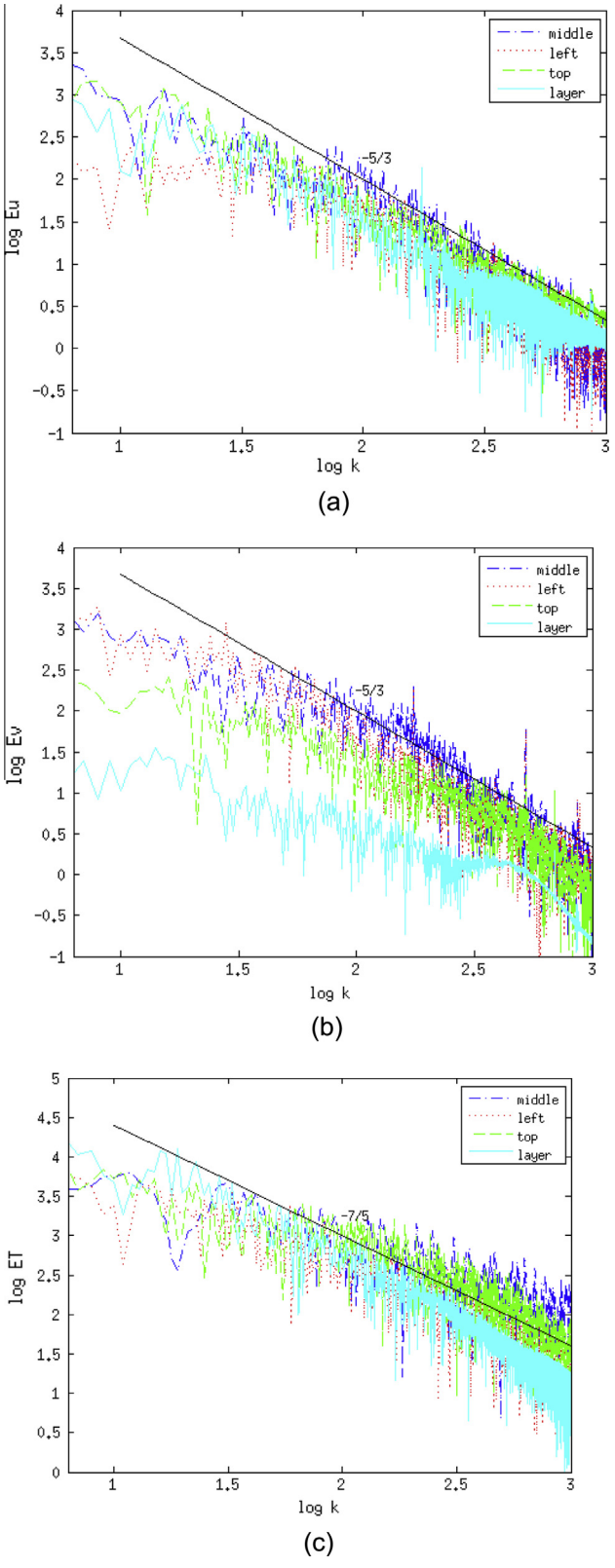


Fig. 4. Kinetic and thermal energy spectra at four positions. (a) Kinetic energy spectrum in *X* direction. (b) Kinetic energy spectrum in *Y* direction. (c) Temperature spectra

4. Conclusions

In this paper, a simple lattice Boltzmann thermal model has been developed to simulate the turbulence Rayleigh–Bénard thermal convection at a high *Rayleigh* number. The present model has been validated by simulating four different cases and comparing the results with those of other methods. The major finding can be summarized as follows:

The numerical evidences showed the good capability of the present method for simulation of turbulence Rayleigh–Bénard convection. The results of simulating show that this simple model is able to calculate turbulence Rayleigh–Bénard convection in the current simulation ranges of $10^6 \leqslant Ra \leqslant 10^{12}$.

The Nu and Ra are linear in logarithmic scale which is in agreement with DNS and empirical value.

Interaction of the thermal plumes has been presented. It can be seen that spectra of the velocity and temperature are in good agreement with the *BO59* scaling.

Considering the intrinsic parallelity of the LBM, this scheme may have good prospects for the simulations of turbulence thermal convection. Even though the results reported in this paper are for two-dimensional cases, it could be useful to find similar scaling laws in three dimensions and it might be important to modify parametric sets in simulations. The latter, allowed us to assess also the importance of small scale thermal plumes on bottom boundary layer for turbulence Rayleigh–Bénard convection.

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